MIL-W-6370C
15 JULY 1963
SUPERSEDING
MIL-W-6370B
29 AUGUST 1957

MILITARY SPECIFICATION

WIRE, ELECTRICAL, INSULATED, ANTENNA

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 **Scope.** This specification covers insulated wire for use in external aircraft antennas.

1.2 **Classification.** The insulated wire shall be of three types determined by differences in conductor material, as specified (see 6.2):

Type I — Copper-covered steel, Part No. 6370-1.

Type II — Copper-covered steel, Part No. 6370-2.

Type III — Beryllium copper, Part No. 6370-3.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-390 — Plastic, Molding Material, Polyethylene, Low and Medium Density

NN-P-515 — Plywood, Container Grade

UU-P-271 — Paper, Wrapping, Waterproofed Kraft

PPP-F-320 — Fiberboard; Sheet, Stock and Cut Shapes

STANDARDS

FEDERAL

FED-STD-141 — Paint, Varnish, Lacquer and Related Materials, Methods of Inspection, Sampling and Testing.

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-109 — Inspection Terms and Definitions

MIL-STD-129 — Marking for Shipment and Storage

(Copies of applicable documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6145

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:58Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

ASTM D1868-61T Method for Corona Measurement (Tentative)

(Application for copies of the standard should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

3. REQUIREMENTS

3.1 Preproduction. The antenna wire furnished under this specification shall be a product which has been tested and has passed the preproduction tests specified herein.

3.2 Materials. Materials shall conform to applicable specifications and shall be as specified herein.

3.3 Construction. The antenna wire shall consist of beryllium copper or copper-covered steel tightly covered with a polyethylene insulator.

3.4 Physical properties.

3.4.1 *Conductor.* The conductor for Type I wire shall consist of extra-high-strength, hard drawn, 40-percent conductivity, copper covered steel wire. The conductor for Type II wire shall consist of high-strength, hard-drawn, 40-percent conductivity, copper covered steel wire conforming to Specification QQ-W-345. The conductor for Type III wire shall consist of 40-percent conductivity beryllium-copper wire. The copper covering of the copper covered steel wire shall be continuous, uniform, and continuously electroplated or welded to the steel core. Each length of conductor shall be continuous, shall be free of joints, and shall be free from lumps, kinks, splints, abrasions, scraped or corroded surfaces, and surface impurities.

3.4.1.1 *Conductor diameter.* The conductor shall have a diameter of 0.0508 ± 0.0010 inch (AWG No. 16).

3.4.1.2 *Breaking strength.* The breaking force or load for each type of conductor shall be as follows:

Type I — 450 pounds minimum.

Type II — 250 pounds minimum.

Type III — 250 pounds minimum.

3.4.1.3 *Elongation.* The elongation in 10 inches for each type of conductor shall not exceed that amount specified below:

Type I — 2 percent

Type II — 5 percent

Type III — 5 percent

3.4.2 *Dielectric covering.* The conductor shall be tightly covered with high molecular weight virgin polyethylene dielectric applied tightly and concentrically to the conductor so that it shall be substantially free from voids to the extent of complying with all the requirements specified herein.

3.4.2.1 *Dielectric material.* The dielectric shall conform to Specification L-P-390 except that a dark brown pigment and an antioxidant (for stabilizing in sunlight) shall be added. The dissipation factor shall be not more than 0.0010.

3.4.2.2 *Dielectric diameter.* The outside diameter of the finished antenna wire shall be 0.183 — 0.004 inch + 0.01 inch.

3.4.3 *Eccentricity.* Unless otherwise specified, the conductor shall not be off center more than 0.02 inch.

3.5 Electrical resistance. The d-c resistance of the conductor of each type wire shall not exceed 11.1 ohms per 1,000 feet at 20° C.

3.6 Performance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:58Z
Check the source to verify that this is the current version before use.

3.6.1 *High voltage.* The completed wire shall withstand the application of a 25,000 volts d.c. potential, or a 60 cycle a.c. potential of 8000 volts rms, between the conductor and the outer surface of the dielectric material for 1 minute without puncturing both before and after 144 hours of accelerated weathering.

3.6.2 *Corona extinction voltage.* The minimum voltage at which corona exists shall be 7,000 volts rms at 60 cps both before and after 144 hours of accelerated weathering, and also before and after the temperature cycling test.

3.6.3 *Insulation shrinkage.* The dielectric covering shall be applied in such a way that subjection to a temperature cycle consisting of a 30-minute exposure at +55° C followed by a 30-minute exposure at —55° C, and to accelerated weathering, shall not result in the exposure of more than 0.20 inch of conductor at either end of a 10-foot length of finished wire.

3.7 Identification of product. Each length of antenna wire shall be permanently marked at approximately 10-foot intervals with the name or symbol of the manufacturer and the part number (see 3.7.1).

3.7.1 *Wire designation.* All wire shall be designated with the appropriate part number as follows:

Type I — Part No. 6370–1

Type II — Part No. 6370–2

Type III — Part No. 6370–3

3.8 Workmanship. The antenna wire shall be manufactured in accordance with the best electrical manufacturing practice.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection and testing of the wire shall be classified as follows:

(a) *Preproduction inspection (does not include preparation for delivery).* Preproduction inspection shall be performed after the award of contract on wire representative of the production item, to determine that the item meets all the requirements of this specification.

(b) *Acceptance inspection.*

(1) Inspection of product for acceptance (see 4.4).

(2) Inspection of preparation for delivery (see 4.7).

4.3 Preproduction inspection.

4.3.1 *Sampling procedures.* Preproduction inspection samples shall consist of 250 feet of wire. Samples shall be identified as required. The wire shall be inspected at a testing laboratory approved by the procuring activity, or at the contractor's plant under surveillance of the procuring activity. (See 6.3.)

4.3.2 *Inspection.* The Preproduction inspection of the wire shall consist of all the tests of this specification as described under 4.6, "Test Methods", and may, at the option of the procuring activity, be supplemented by tests under actual service conditions com-

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:58Z
Check the source to verify that this is the current version before use.

patible with the requirements specified herein.

4.4 **Inspection of product for acceptance.** Inspection of product for acceptance shall consist of groups A, B, and C inspection. These tests shall be performed before packaging.

4.4.1 *Inspection lot.* For groups A and B inspection, an inspection lot shall be the sum of the number of individual pieces contained in the reels submitted for Government acceptance (see 5.1.1).

4.4.2 *Resubmitted lots.* If an inspection lot is rejected, the manufacturer may rework the lot or screen out defective lengths and resubmit it for Acceptance inspection. Resubmitted lots shall be kept separate from new lots. The resubmitted lots shall be inspected using tightened inspection. Before an inspection lot is resubmitted, full particulars concerning the cause of previous rejection and the action taken to correct the defects found in the inspection lot, shall be furnished by the contractor to the Government inspector.

4.4.3 *Group A inspection.* Group A inspection shall consist of the examination and tests specified in table I. Statistical sampling and inspection shall be in accordance with the ordinary inspection procedure of Standard MIL-STD-105.

TABLE I. *Group A inspection*

Inspection	Requirement paragraph	Inspection paragraph	AQL major	AQL minor
Examination of product	3.2, 3.3, 3.4, 3.4.1, 3.4.1.1, 3.4.2, 3.4.2.1, 3.4.2.2, 3.4.3, 3.7.1, and 3.8.	4.6.1	2.5 percent	10 percent
High voltage	3.6.1	4.6.2	1.0 percent	

4.4.4 *Group B inspection.* Group B inspection shall consist of the Electrical resistance test as specified in 4.6.4 and shall meet the requirements of 3.5.

4.4.4.1 *Sampling procedure.* The sampling procedure shall be in accordance with the special procedures for small sample inspection of Standard MIL-STD-105. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1. The AQL shall be 6.5 percent defective, the inspection level for normal and tightened inspection shall be L-7, and for reduced inspection L-5.

4.4.4.2 *Disposition of sample units.* Sample units which have passed the Group B inspection shall be delivered on the contract or order, if the lot is accepted.

4.4.5 *Group C inspection.* Group C inspection shall consist of the tests specified in Table II.

TABLE II. *Group C inspection*

Inspection	Requirement paragraph	Inspection paragraph
Breaking strength and elongation	3.4.1.2, 3.4.1.3	4.6.3
Corona extinction voltage	3.6.2	4.6.5
Accelerated weathering	3.6.3	4.6.6
Temperature cycling	3.6.3	4.6.7

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:58Z
Check the source to verify that this is the current version before use.

4.4.5.1 *Sampling for group C inspection.* A 10-foot sample shall be selected at random from each 10,000 feet in the lot or fraction thereof on the order and subjected to the tests specified in 4.4.5. These tests shall be in addition to the group A and group B inspection. If this sample fails to meet any of the requirements other than the corona extinction voltage, two additional samples shall be chosen from the lot and tested. If either of these samples fails, or if the original 10-foot sample fails the corona extinction voltage test, each length of finished wire of the lot shall be sampled and tested as the basis for acceptance or rejection.

4.5 **Definitions.** Standard MIL–STD–109 applies for definitions of inspection terms used herein.

4.6 Test methods.

4.6.1 *Examination of product.* Each sample piece of antenna wire shall be examined to determine conformance with this specification with respect to material and workmanship. Sufficient spot checks shall be made to assure conformance with the diameter tolerance, minimum insulation wall thickness, and eccentricity requirements of this specification.

4.6.2 *High voltage.* Each sample length of wire shall be subjected to a 25,000-volt DC, or an 8000-volt r.m.s. at commercial power frequency, potential between the conductor and the outer surface of the dielectric for a period of not less than 1 minute. This test may be conducted with the wire in a conducting bath or close-fitting conductor which serves as one electrode of the high-voltage circuit.

4.6.3. *Breaking strength and elongation.* Three full cross-section specimens shall be marked with a 10-inch-gage length and tested for compliance with the breaking strength and elongation requirements of this specification. The testing machine shall be sensitive to a variation of 0.4 percent of any recorded load and shall be calibrated when required by the Inspector. The speed of the testing machine shall be such that the load is accurately indicated at all times. Specimens breaking outside of the gage marks shall be disregarded and additional specimens tested.

4.6.4 *Electrical resistance.* The wire shall be tested for direct-current resistance while at an ambient temperature of 20° C ± 2° C or, if another temperature is used, an approved temperature correction shall be applied.

4.6.5 *Corona extinction voltage.* The corona extinction voltage shall be determined by the methods described in ASTMD 1868–61T. This test may be conducted with the wire in a conducting bath or close-fitting conductor which serves as one electrode of the high-voltage circuit. Corona at the ends of this outer conductor may be suppressed by oil immersion or other means.

4.6.6 *Accelerated weathering.* A 10-foot length of antenna wire shall be exposed in a single layer coil, 18 inches in radius, to the light from a carbon-arc source. The source shall be the same as that described in method 6152 of Standard FED–STD–141, or an approved equal, except that the temperature within the apparatus shall be 55° C $\pm^{3}_{0}$ C. After exposure for 144 hours, the polyethylene shall show no visible evidence of crazing or cracking, and the shrinkage of the polyethylene along the conductor shall not exceed 0.20 inch. Shrinkage shall be measured in accordance with the measuring and marking procedures of 4.6.7. After exposure to this test, the wire shall satisfactorily meet the requirements of 3.6.1 and 3.6.2.

4.6.7 *Temperature cycling.* The test to determine the amount of dielectric deformation owing to temperature cycling shall be accomplished by placing a 10-foot length of antenna wire in a test chamber. The dielectric covering shall be stripped off to expose

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:58Z
Check the source to verify that this is the current version before use.

2 inches of conductor at each end of the test sample, and the conductor shall be marked with paint, ink, or dye in order that the initial boundary of the dielectric will be plainly identified at the start and end of the test. The sample shall then be subjected to a 30-minute exposure at + 55° C. $^{+5°}_{-0°}$ C. followed by a 30-minute exposure at — 55° C. $^{+0°}_{-5°}$ C. The change of temperature between these two exposures shall be accomplished within 20 minutes. Within 10 minutes of the termination of the exposure at — 55° C, the sample shall be removed from the chamber and be subjected to a 75-pound tensile stress, applied at the ends of the conductor, and the amount of bare conductor measured. The distance between the originally marked dielectric boundary and the post-cycling boundary shall not exceed 0.20 inch at either end of the samples.

4.7 Acceptance inspection of preparation for delivery. Preparation for delivery shall be inspected to determine conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A.* Unless otherwise specified, the wire shall be wound on sturdy reels of suitable design to protect the wire from injury in handling, rolling, stacking, and storage. The diameter of the core of the reel shall be not less than 10 inches. Reels shall contain a total of 4,000 to 6,000 feet of wire in not more than six lengths; the shortest length per piece shall be not less than 100 feet.

5.1.2 *Level C.* Wire shall be preserved and packaged in accordance with manufacturer's commercial practice.

5.2 **Packing.**

5.2.1 *Level A.* Wire preserved and packaged as specified in 5.1.1 shall be protected from abrasion or damage by the use of plywood, solid fiberboard, waterproof barrier material, or equivalent material cut to the full inside width of the reel drum and wrapped over the wire around the drum. The wrapper shall be tied with wire or metal banding. When used, plywood shall conform to Specification NN-P-515, type I or type II, class 2. Solid fiberboard, when used, shall conform to Specification PPP-F-320, Type SF, Class-weather resistant, Grade V3S. Waterproof barrier material, if used, shall conform to Specification UU-P-271.

5.2.2 *Level C.* Packages which require overpacking for acceptance by the carrier shall be packed in exterior-type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall meet Consolidated Freight Classification Rules or regulations of other common carriers as applicable to the mode of transportation.

5.3 Marking of shipment. Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129. The identification shall be composed of the following information listed in the order shown:

Stock No. or other identification number as specified in the purchase document*

WIRE, ELECTRICAL, INSULATED, ANTENNA

Part No. ..... Type .....

*NOTE: The contractor shall enter the Federal Stock No. specified in the purchase document or as furnished by the procuring activity. When the Federal Stock No. is not provided or available from the procuring activity, leave space therefor and enter the Stock No. or other identification when provided by the procuring activity.

6. NOTES

6.1 Intended use. The wire covered by this

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:58Z
Check the source to verify that this is the current version before use.

specification is intended for use in external aircraft antennas to limit the starting of a corona discharge upon buildup of high voltage, and thereby reduce radio interference of precipitation static origin.

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type of wire desired.

(c) Quantity of wire required.

(d) Level of packaging and packing required.

6.3 **Provisions for preproduction tests.** Manufacture of any further wire on the particular contract shall be suspended until the samples submitted are pronounced satisfactory by the procuring activity. When a contractor is in continuous production of the wire from contract to contract, submission of further preproduction samples on the subsequent contracts may be waived at the discretion of the procuring activity. Approval of preproduction samples or the waiving of preproduction tests does not preclude the requirements of acceptance testing.

6.3.1 It shall be understood that the wire supplied under contract or order shall be identical to the corresponding preproduction sample in design, construction, quality, material, workmanship, and method of manufacture. Deviation from the standards of the preproduction sample shall be made only by the procuring activity. Evidence of unauthorized change shall constitute cause for rejection.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Electronics Command
Navy—Wep
Air Force—ASD

Preparing activity:
Navy—Wep

(Project 6145-0196)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:58Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 13413

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:58Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:58Z
Check the source to verify that this is the current version before use.